

Khalil LAGHA

M2 student — Electrical Energy Systems Design (CSEE) · Université Grenoble Alpes

Ranked 2nd of the M1 EEA cohort (15.43/20). DC-DC converters (buck, boost), regulation-loop tuning, electrical machines and power grids — from the lab (GIPSA-lab) to the industrial field. Work-study rotation: 2–3 weeks company / university.

- Grenoble, France

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Authorized to work in France

01 Experience

- Research Intern · GIPSA-lab, Grenoble

04 – 07/2026 · 3.5 months

Developed a Matlab/Simulink model of a PEMFC fuel cell — 6% dynamic / 1% static error.

Tuned the regulation loop of a **boost DC-DC converter** (PI controller), targeting conversion-efficiency optimization — validated in simulation then on a real test bench.
- Intern · SLB (Schlumberger)

07/2025 · 1 month

Wrote technical syntheses in an international industrial environment; earned the **NEST & SIPP Level 2** HSE certifications.
- Power Electronics & Automation Intern · EURL Lagha

01/2024 · 15 d + 09/2024 · 1 month

Contributed to an industrial **buck DC-DC converter** — 50 V input, adjustable 0–48 V output, **25 A**.

Worked on **4 transformer-rectifiers** for pipelines: 3-phase 400 V → 100 V, rectification + filtering (ripple < 5%, EMI compliance) — output **100 V DC / 80 A (≈ 8 kW)**.

Co-designed the supervision HMI (Siemens TIA Portal); produced the electrical schematics in EPLAN.
- Power Grid Intern · Sonelgaz

03/2024 · 15 d · observation

Studied substation structure and the full generation → distribution chain (**400/220 kV → 60 kV → 30/10 kV → 400/230 V**); visited an MV/LV substation.

02 Projects

- Power grid analysis · Matlab · solo · github.com/KhalilEllahLAGHA

Load flows from **3 to 118 buses** (IEEE 14/30/57/118) by Gauss-Seidel, Newton-Raphson and fast-decoupled; direct Zbus = inv(Ybus) to 1e-12; critical fault-clearing-time study in transient stability.
- Winding and flux analysis program · Python / PyQt · FEMM

Three-phase winding, tooth-based MMF, reluctance-network flux density, finite-element validation.
- Machine control · Matlab/Simulink

FOC of an induction motor (Park, PID, observers); cascade control of a DC motor through a three-phase thyristor bridge (outer speed loop, inner current loop).

2nd

of the M1 EEA cohort (UGA)

15.43

M1 annual average /20

C2

English

03 Education

- M2 CSEE — Electrical Energy Systems Design

2026 –

Université Grenoble Alpes · work-study
- M1 EEA (EECS)

2025 – 26

Université Grenoble Alpes · 2nd in class
- Engineering cycle, Electrical Engineering

2021 – 25

École Nationale Polytechnique d'Alger · 4 of 5 years completed; prep classes: top-ranked
- Baccalauréat in Mathematics

2021

Highest honours — 16.93/20

04 Skills

- Power electronics

DC-DC / AC-DC conversion
- Machines & transformers

Grids: load flow, stability

Loop tuning (PID, LQR)

PLC / HMI / SCADA

Electrical CAD: EPLAN, AutoCAD

AI: NN, GA, fuzzy logic

05 Software

Matlab/Simulink · Python · C/C++ · LTspice · Proteus · EPLAN · AutoCAD Electrical · Siemens TIA Portal · Step7 · SolidWorks · Arduino

06 Languages & certification

- French

C1
- English

C2
- Arabic

native
- SLB NEST & SIPP Level 2 — 07/2025 (HSE)

07 Soft skills

- Fast learner

Rigorous

Organized
- Problem solving

Comfortable with tools

Portfolio & projects: khalilellahlagha.github.io · IEEE Student Branch ENP · VIC